

Event-Driven Programming



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About Me

- Bachelor's degree in Mathematics & Computer Science
- Master's degree in Computer Science - Software Engineering
- Over 12+ years of experience in **Enterprise Software Development & Training**
- Worked across Healthcare, Banking, Education, and Digital Marketing Space
- Member of Computer Professionals (Registration Council of Nigeria) [CPN]
- Member of Google Developer Group (GDG) & Community Mentor
- Microsoft Certified Azure Developer Associate (AZ-204) & Community Mentor
- Claude Certified Solutions Architect & Community Mentor



Overview

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**Role of
JavaScript**

1

**JavaScript
Fundamentals**

2

**BOM BOM
& DOM**

3

**Events Triggers
& Handlers**

4

**Coding Session
w/Northwind AI**

Prerequisites

- ☐ HTML Fundamentals
- ☐ CSS Fundamentals
- ☐ Charged Laptop
- ☐ Internet Connection



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Role of JavaScript

Webpage Interactivity

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JavaScript is an **interpreted**,
high-level programming language
used to develop interactive webpages.

Trivia

- ❑ **1995** - JavaScript was created by Brendan Eich (Netscape, *within 10 days*).
- ❑ **2009** - Node.js was created by Ryan Dahl as a JS runtime environment to execute JS outside web browsers.
- ❑ **2011** - React.js was created by Jordan Walke (Facebook) as an open-source JS library for building single-page applications (SPAs).

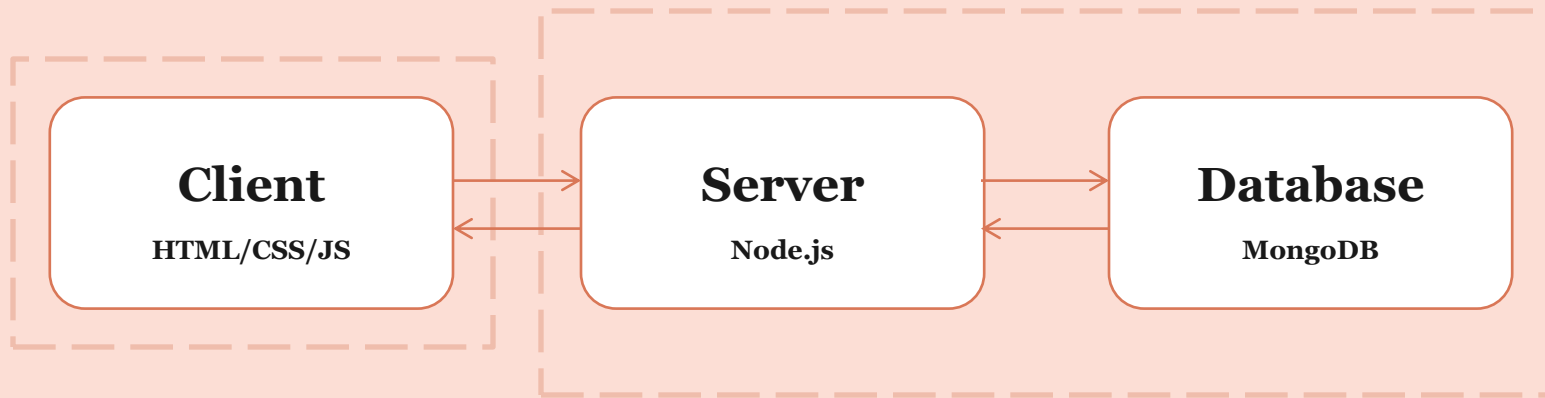


Application Areas

- ❑ Manipulating HTML & CSS (DOM Elements)
- ❑ Handling Webpage Interactions
- ❑ Building APIs/Web Services (Node.js)
- ❑ Building Desktop Apps (Electron)
- ❑ Building Mobile Apps (React Native)
- ❑ Interfacing with Ethereum Smart Contracts (Web3.js)



Web App Architecture





Questions

1

JavaScript Fundamentals

Syntax & Semantics

- ❑ **Variables & Constants** – Named storage location in memory that can/cannot be changed after declaration (var, const).
- ❑ **Data Types** – Defines the type of value to be stored in memory, how much memory it consumes, and what type of operations can be performed on it;
 - ❑ Primitives (Number, String, Boolean, null, undefined)
 - ❑ Non-Primitives (Array, Object, Date)
- ❑ **Operators** – Symbols used to perform computations;
 - ❑ Arithmetic
 - ❑ Assignment
 - ❑ Comparison
 - ❑ Logical
 - ❑ *Ternary

Variables & Constants

1.1

Example 1

1. `// TODO: Compute Student Grade`
2. `var score = 47;`
3. `const total = 60;`
4. `const grade = (score * 100) / total; // 78.33%`
5. `const passed = grade >= (total / 2); // true`
6. `const remarks = passed ? "Pass" : "Fail";`

Identifiers

1. The name given to a variable is known as an **Identifier**.
2. Identifiers can contain letters, digits, underscores, and dollar signs.
3. Identifiers can begin with a letter, a dollar sign (\$) or an underscore (_).
4. Numbers cannot be used as the first character of an Identifier.
5. The JavaScript convention for Identifier is *camelCase.
6. Identifiers should be as descriptive as possible. This helps with readability.

Example 2

1. `// TODO: Compute Person Age`
2. `var birthYear = 1952; // BAT ☐`
3. `const thisYear = new Date().getFullYear(); // 2026`
4. `const age = thisYear - birthYear; // 74`
5. `const ageGroup = "Senior man ☐ ";`
6. `const canVote = age >= 18 ? "Sure" : "Nope";`

Comments

- ❑ Comments are primarily used to document code snippets.
- ❑ Comments can also be used to ignore code snippets during debugging.
- ❑ Inline comments begin with `// ...`
- ❑ Multi-line comments are wrapped within `/* ... */`

Data Types

1.2

Array

- ❑ An array is an ordered list of values known as **elements**.
- ❑ Array elements are accessed based on their **index**.
- ❑ The first value in an Array is located at index 0.
- ❑ The last element in an Array is located at index $n - 1$, where n is the size of the Array.
- ❑ Arrays can grow or shrink as elements are added or removed.
- ❑ Arrays can store elements of different data types (numbers, strings, booleans, etc).
- ❑ A nested Array is called a **Matrix**.

Example 1

```
1.  //TODO: Process Jamb Scores
2.  const jambScores = [56, 42, 55, 60]; // ENG, MAT, PHY, ECO
3.  var jambScoresOK = Array.isArray(jambScores); // true
4.  var size = jambScores.length; // 4
5.  var english = jambScores[0]; // 56
6.  var economics = jambScores[size - 1]; // 60

7.  var removedLastValue = jambScores.pop(); // [56, 42, 55], 60
8.  var newArraySize = jambScores.push(55); // [56, 42, 55, 55], 4
9.  var removedFirstValue = jambScores.shift(); // [42, 55, 55], 56
10. var newArraySize = jambScores.unshift(76); // [76, 42, 55, 55], 4
11. var scored_56 = jambScores.includes(56); // false
12. var scored_76 = jambScores.includes(76); // true
13. var sortedScoresAsc = jambScores.toSorted((a, b) => a - b); // [42, 55, 55, 76];
14. var sortedScoresDesc = jambScores.toSorted((a, b) => b - a); // [76, 55, 55, 42];
15. var minScore = Math.min(...jambScores); // 42
16. var maxScore = Math.max(...jambScores); // 76
17. var totalScore = jambScores.reduce((total, value) => total + value, 0); // 228

18. jambScores.toString(); // [76, 42, 55, 55]
```

1.3

Operators

Arithmetic

Oper	Name / Link	Example
+	Addition	$x = y + 2$
-	Subtraction	$x = y - 2$
*	Multiplication	$x = y * 2$
/	Division	$x = y / 2$
**	Exponentiation	$x = y ** 2$
%	Remainder	$x = y \% 2$
++	Increment	$x = y++$
--	Decrement	$x = y--$

Assignment

Oper	Name / Link	Example	Same As
=	Simple	x = y	x = y
+=	Add	x += y	x = x + y
-=	Subtract	x -= y	x = x - y
*=	Multiply	x *= y	x = x * y
/=	Divide	x /= y	x = x / y
%=	Remainder	x %= y	x = x % y

Comparison

Oper	Name / Link	Comparing	Returns
==	Equal to	x == 8	false
==	Equal to	x == 5	true
===	Strict Equal	x === "5"	false
===	Strict Equal	x === 5	true
!=	Not Equal	x != 8	true
!=	Strict not Equal	x !== "5"	true
!=	Strict not Equal	x !== 5	false
>	Greater than	x > 8	false
<	Less than	x < 8	true
>=	Greater or Equal	x >= 8	false
<=	Less or Equal	x <= 8	true

Logical

Oper	Name / Link	Example
&&	<u>AND</u>	(x < 10 && y > 1)
	<u>OR</u>	(x === 5 y === 5)
!	<u>NOT</u>	!(x === y)
??	<u>Nullish Coalescing</u>	x ?? y



Questions

2

BOM BOM & DOM

Browser/Document Object Model



Questions

3

Events Triggers & Handlers



Questions

4

Coding Session

Northwind AI



Questions

Resources

1. Tugbeh, E. O. (2026). [PDF] Event-Driven Programming:
<https://github.com/2gbeh/src/blob/main/event-driven-programming/slide.pdf>
2. Tugbeh, E. O. (2026). [ZIP] Source Code Repository:
<https://github.com/2gbeh/src/blob/main/event-driven-programming/lab.zip>
3. W3Schools – Introduction to Programming:
<https://w3schools.com/programming/index.php>
4. W3Schools – JavaScript Tutorial: <https://w3schools.com/js/default.asp>

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